Image Segmentation and Morphological Processing

PCD_Assignment03

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1. Aim and example image

The aim of this assignment is to implement a segmentation method and several morphological operations without relying on ready-made image-processing functions. I used Otsu thresholding to divide a flower photograph into two intensity classes, then compared the original binary mask with masks processed using morphology. The main question is how these operations change the shapes, small regions, and gaps in the segmentation.

The example is the supplied flower photo, which contains purple and pink petals, green leaves, dark flower centers, and pale wrapping. Its original size is 242 × 226 pixels, or 54,692 pixels in total. I kept that resolution and did not add artificial noise. The image is stored as images/flowers.png in the repository.

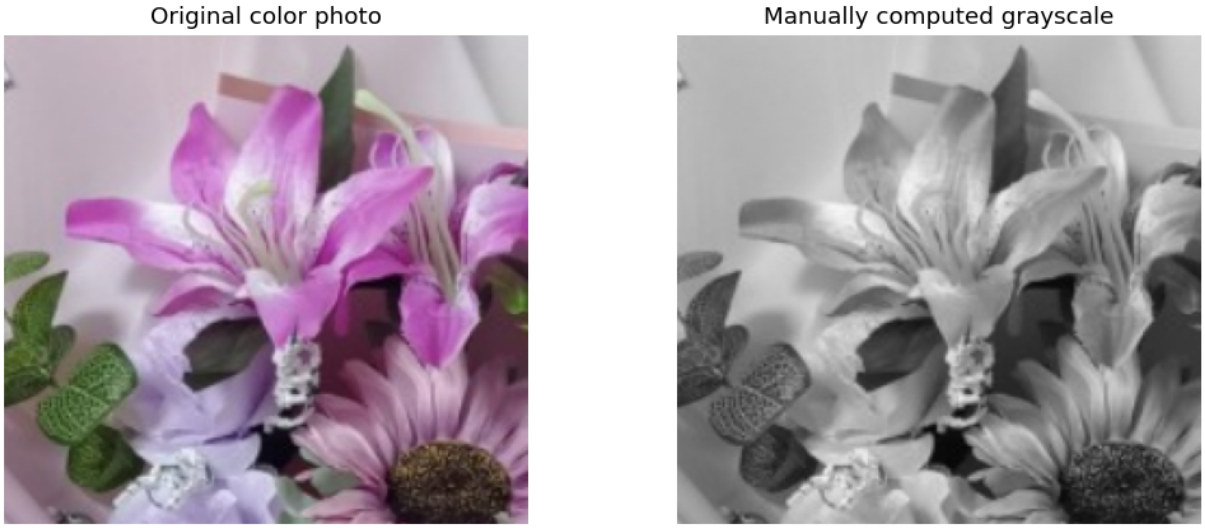


Figure 1. Supplied flower photograph and the grayscale image used for segmentation.

2. Segmentation method

I first converted the color pixels to grayscale using Y = 0.299R + 0.587G + 0.114B. The calculation is written with Python loops. I then built a 256-bin histogram and tested the possible thresholds. Otsu selects the threshold that maximizes the separation between the two intensity classes, measured by their between-class variance.

For this image, the selected threshold was 129. Pixels with intensity greater than 129 became white foreground; the remaining pixels became black background. Here, foreground means the brighter intensity class. It does not mean that every white pixel belongs to a flower.

3. Morphological processing

All treatments start from the same Otsu mask and use a centered 3 × 3 square structuring element. I implemented each neighborhood operation using lists and loops, returning a new mask rather than changing the input. Pixels outside the frame are treated as zero. This border rule matters because several bright regions in the photo touch the image edge.

Operation	**Effect**
Erosion	Keeps a white pixel only when every active neighbor is white; shrinks white regions.
Dilation	Makes a pixel white when any active neighbor is white; expands white regions.
Opening	Erosion followed by dilation; suppresses small white regions and protrusions.
Closing	Dilation followed by erosion; fills small dark gaps and holes.
Morphological gradient	Dilated mask minus eroded mask; gives a band around boundaries.
Inner boundary	Original mask minus eroded mask; keeps the boundary inside white regions.

I compared the individual operations and both composition orders. The main before-and-after example uses opening followed by closing: opening suppresses small white structures, and closing fills small black gaps. I also included the reverse order because the first operation changes the input of the second. The gradient and inner boundary are shown separately in the notebook because they represent edges rather than complete segmented regions.

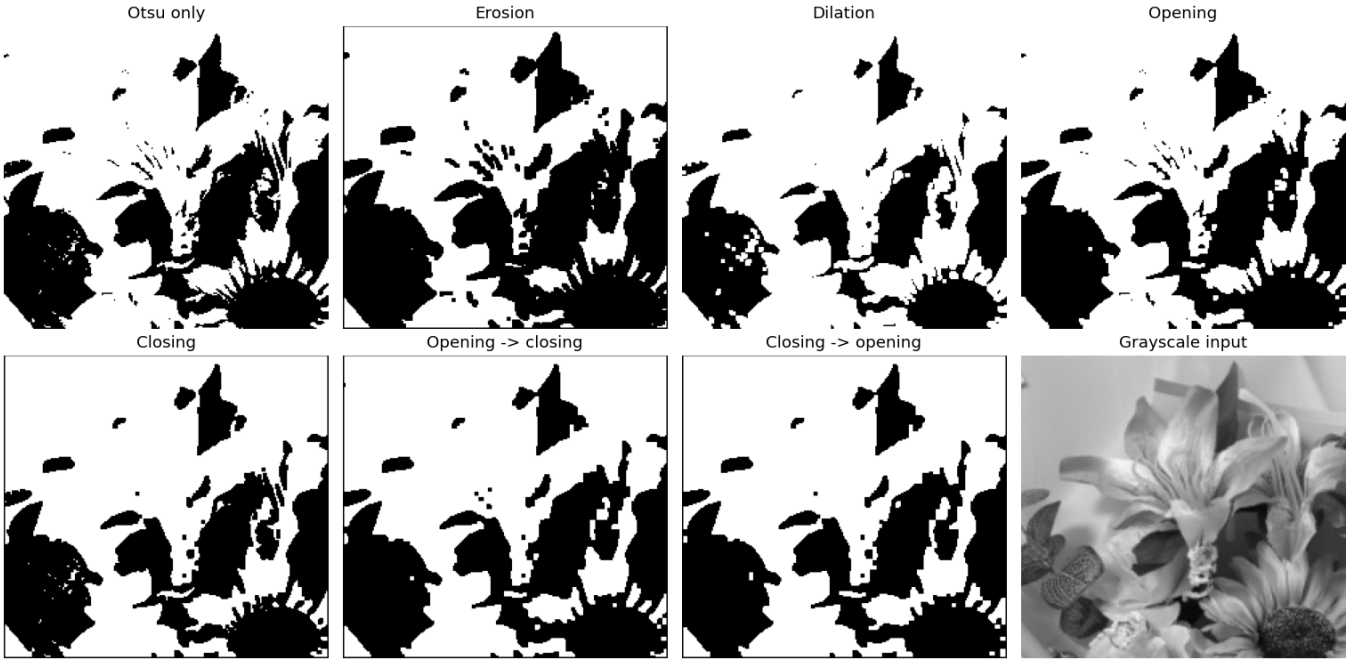


Figure 2. Otsu-only mask compared with the morphological treatments. White is the brighter intensity class.

4. Results and comparison

The table reports foreground size and changes relative to Otsu alone. Added pixels change from black to white; removed pixels change from white to black. These are descriptive counts, not counts of corrected mistakes.

Method	Foreground	Coverage	Added	Removed
Otsu only	34,518	63.11%	0	0
Erosion	29,798	54.48%	0	4,720
Dilation	38,600	70.58%	4,082	0
Opening	33,834	61.86%	0	684
Closing	34,575	63.22%	699	642
Opening → closing	33,605	61.44%	352	1,265
Closing → opening	34,207	62.54%	676	987

Erosion reduced foreground coverage to 54.48%, while dilation increased it to 70.58%. This matches their expected behavior: erosion contracts white regions, whereas dilation expands them. On a flower image, this can remove narrow bright petal details or connect regions that were previously separate.

Opening followed by closing changed foreground coverage from 63.11% to 61.44%. It added 352 pixels and removed 1,265, giving a net decrease of 913 foreground pixels. The two composition orders produced different masks and counts, so their order has a visible effect. Some removed pixels lie along the frame because of zero padding; they should not all be described as noise.

The most noticeable limitation is that the pale background and bright petals overlap in grayscale intensity. Otsu selects parts of both, while darker flower details are excluded. Morphology can alter the local shape of those regions, but it cannot decide which regions are flowers. There is no annotated reference mask for this photo, so I do not report accuracy, IoU, or Dice, or claim that the processed mask is more correct.

5. Conclusion and implementation check

The implementation shows how intensity-based segmentation and morphological processing affect different aspects of an image. Otsu determines the initial division between bright and dark pixels, while morphology changes the spatial structure of that division. The flower example makes the tradeoff clear: small regions and gaps can be changed, but real detail can also be lost. The appropriate operation depends on the intended object and the details that need to be preserved.

The notebook contains 17 hand-computable algorithm checks, plus metric and mask-statistic checks. All ten code cells ran successfully. Pillow is used for image decoding and the original-color display, and Matplotlib is used for figures. Segmentation, grayscale calculations, morphology, and mask statistics use no library image-processing functions.

Repository: https://github.com/yahaiprazu/PCD_Assignment03